

CSC8645- Assignment

Submission Due: 4pm, Friday 20th March '26 | Demonstration: 20th March

Brief

The assignment consists of two main tasks: 1) computer vision and 2) natural language processing. For each task, you will design, implement, and evaluate appropriate models using the provided datasets. You are required to present your results and submit a structured technical report. The report should include (for each task separately)

1. a brief introduction to the problem (you are encouraged to cite a couple of current literature),
2. description of your approach (methodology) including flowchart(s),
3. presentation of the results,
4. key findings from the results and associated discussions and
5. conclusions.

You need to demonstrate (**compulsory**) the code and answer related questions on designated days. If you fail to demonstrate (/absent), the entire corresponding task(s) will be marked as zero.

The submission should include:

1. A PDF report (a single file for both tasks)
2. A .zip file containing Python codes (separate files for two tasks) in Jupyter Notebook format.

TASK 1: Food identification with segmentation

Food management is an important part of human well-being and care. Dietary intake monitoring is important for the overall well-being of patients in hospitals and residents at care homes. The aim of this assignment is the implementation of an intelligent food recognition system. Your task is to design and develop your own algorithm to semantically **segment** individual food items on a plate.

Specifically, you are asked to

1. Design and development of your own segmentation algorithm. You may also make use of any pre or post processing steps. While designing the algorithm you may want to
 - a. Get inspiration from existing literature (one or more) which you must cite in your report and code.
 - b. Design the baseline algorithm and make further improvement in consecutive stages.For demonstration purposes the input to your code will be an image with food items (chosen by you ahead of the session). The output will be segmented image and scores of comparison metrics. **[25]**
2. Evaluation of segmentation can be done by comparing the segmentation output with the ground truth for given images on measurable metrics, such as, Mean Intersection over Union (mIoU). You are expected to compare the performance using this metric. **[5]**
3. Generate and report results with graphs/tables (and sample images) using the given dataset original and ground truth data¹. **[10]**
4. Write a report as described at the beginning of the brief. **[10]**

¹ <https://xiongweiwu.github.io/foodseg103.html>

TASK 2: Multimodal Idiomaticity Representation

Task Context: Which of these images best represents the meaning of the phrase bad apple in the following sentence?



"We have to recognize that this is not the occasional bad apple but a structural, sector-wide problem"

How about here?

"However, if ethylene happens to be around (say from a bad apple), these fruits do ripen more quickly."

Even if you aren't already familiar with the non-literal idiomatic meaning of *bad apple* (an individual whose influence negatively impacts the behaviour or reputation of a group or organisation), as a human you might be able to answer this question fairly easily.

Computational language models, on the other hand, struggle with figurative expressions such as these.

By using visual representations like these, this task aims to push participants to improve the quality of model representations of idiomatic expressions and develop models which come closer to "understanding" the semantic meaning of idioms, which are an important feature of natural language.

You will be presented with a set of 3 images and a context sentence in which a particular potentially idiomatic nominal compound (NC) appears. The goal is to predict which image is correctly represent the sense in which the NC is used in the given context sentence. For example,

Input:

- Compound: "Bad Apple"
- Context Sentence: "The problem for them, of course, is how to explain how these few bad apples managed to stay in place for so many years."
- 3 Candidate Images with captions (as separate text):



Output: *Predict which image is the correct image for the given compound and context sentence.*

Data: English training data for both subtasks can be obtained from the canvas page. It has 3 subfolders: train, dev and, test. For each subfolder, you will get a “.csv” file that contains the detailed information including compound, context sentence, image id, image caption and label and a subfolder “images” that contains all the images.

You are free to explore different techniques such as using NLP analysis, pretrained embeddings, fine-tuning approaches, contrastive learning, and using LLMs to augment data, etc. You must use appropriate performance metrics including Accuracy and F1 score to evaluate the outcome of your predictions. To begin the work, a skeleton code is provided for you to build upon: <https://colab.research.google.com/drive/1mD8pw1U3qO6IfSMOSpMBQgH4qb5TFXZo?usp=sharing>.

Requirements and Grading Criteria

1. **NLP Exploration Data Analysis:** How well have you explored the data and how it helps with the rest of the work [5]
2. **Methods:** How well have you developed your pipelines [25]
 - a. LLM based Data Augmentation
 - b. Zero-shot prediction
 - c. Fine-tuning based and other prediction approaches
3. **Results and analysis:** How good are the results and how well have you analysed them [10]
4. **Report:** How well have you reported and discussed what you have done and explained why you did things [10]

DEMONSTRATION

Demonstrate your code and answer questions for both tasks (10 minutes). For both tasks, you need to make sure your code is working and have an input data chosen prior to the session for testing your models. [pass/fail]

Total mark for the ASSIGNMENT: [100]

NOTE 1: You can use suitable main library functions for the tasks.

NOTE 2: Reference paper relating to the dataset in task 1: Wu, Xiongwei, et al. "A large-scale benchmark for food image segmentation." Proceedings of the 29th ACM International Conference on Multimedia. 2021. <https://dl.acm.org/doi/pdf/10.1145/3474085.3475201>

Submission

The assignment is worth 100% of the overall grade for CSC8645 and is compulsory. The report (including both tasks) should be in the order of **1,000** words in length (approximately two pages long), excluding figures, tables and references. Please refer to the “Brief” section for the type of submissions.

Referencing

You must follow the IEEE referencing style details which can be found here:

<https://ieeauthorcenter.ieee.org/wp-content/uploads/IEEE-Reference-Guide.pdf>

Plagiarism

Work which is submitted for assessment must be your own work. Plagiarism means presenting the work of others as though it were your own. Further details about the university policy on plagiarism can be found here:

<https://www.ncl.ac.uk/academic-skills-kit/good-academic-practice/plagiarism/>

Marking and feedback

Marks will be given for all tasks and sub-tasks. Please note that the marking will not be linear, which means achieving higher grades will be increasingly challenging and should meet the expectations (and beyond) of the marking scheme.

Written feedback will be provided along with the marks within 20 working days of the submission deadline. If you are not satisfied with the marks and feedback, you are welcome and encouraged to discuss individually with the module team.
